HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Engineered and supported **automation** systems for production lines, directly contributing to improvements in **quality**, **profitability**, and **safety**.
- Developed and executed **project assignments**, ensuring compliance with timelines and established governance policies, while meticulously maintaining required **project management documents**.
- Performed **design work**, including layouts and concepts, to support new projects and optimize existing **plant layouts**.
- Evaluated and maintained **machine safeties** for automation equipment, ensuring employee protection during operation.
- Analyzed **production data** to ensure automation lines met targeted output and OEE metrics, maintaining finished product **quality**.
- Troubleshot and repaired **automation** equipment, including assembly machines, fixtures, label applicators, and robots.
- Containerized **Java Spring Boot** applications using **Docker** and deployed them to **AWS EC2** instances, managing application configurations and resource allocation.

SKILLS

Manufacturing Engineering – Automation, Production Lines, Quality, Profitability, Safety, Machine Safeties, OEE, Assembly Machines, Robots, Test Equipment.

Project Management – Project Assignments, Timelines, Governance Policies, Project Management Documents, Design Work, Layouts, Concepts, Engineering Change Requests.

Software Development – Java, Spring Boot, REST APIs, Docker, AWS, EC2, S3, Lambda, CloudWatch, PostgreSQL, MySQL, Oracle.

Data Analysis – Production Data Analysis, Data Validation, SQL.

Operations Support – Incident Response, SLA Management, Change Management, CI/CD, Jenkins, Git, Jira, Confluence, Linux.

Data Visualization – Power BI, Tableau, DAX, Power Query.

PROJECTS

Industrial Automation Enhancement Project

- Designed and implemented a new automated assembly fixture for a critical production line, improving throughput by **18%** and reducing manual handling defects.
- Developed Python scripts to parse and analyze sensor data from multiple automation units, identifying key failure patterns that led to a **22%** reduction in unscheduled downtime.
- Documented the complete automation workflow, including safety protocols and maintenance procedures, for use by operations teams and new engineers.

Production Data Analytics Dashboard

- Built an interactive **Power BI** dashboard that visualizes key production metrics, including OEE, output, and defect rates, enabling real-time monitoring for stakeholders.
- Developed **SQL** queries across PostgreSQL and Oracle databases to extract and transform production data, ensuring data integrity for reporting.
- Created custom **DAX measures** to calculate complex performance indicators, providing actionable insights into process efficiency.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)